Team Problems

1. Compute the number of digits in the base-10 representation of $2^3 3^4 4^5 5^6$.

2. Compute the length of the interval of the set of numbers x that satisfy $\lfloor x^2 \rfloor = \lfloor (x+3)^2 \rfloor$, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x.

3. The sequence $x_1, x_2, x_3, \ldots$ is formed by setting $x_1 = 1$, $x_2 = -3$, and for $n > 2$, x_n is obtained by cubing one of the previous terms, chosen uniformly at random. Compute the probability (as a fraction) that $x_1 x_2 x_3 x_4 x_5 > 1000$.

4. Compute the remainder when $8^8 - 4^4$ is divided by $4^4 - 2^2$.

5. A, B, and C are chosen at random (with replacement) from the set of digits $\{0, 1, 2, \ldots, 9\}$. Compute the probability (as a fraction) that the units digit of A^{B^C} is a six. Note that 0^0 is undefined.

6. Compute the smallest integer b with exactly 4 positive integer divisors such that, when expressed in base b, the number $b!$ ends in exactly 10 zeroes.

7. Three balls of radius 1 are sitting on the surface of a table and are mutually tangent to each other. A solid circular cone of height 2 has its base on the surface of the table, and the lateral surface of the cone is tangent to each of the three spheres. Compute the radius of the base of the cone.

8. Compute the smallest positive angle x in degrees such that $\cos^2 x = \dfrac{1 - \sin 46^\circ}{2 \sin^2 112^\circ - \sin 46^\circ + 1}$.

9. For non-zero complex numbers a and b, the chain of equalities $\dfrac{a}{b+1} = \dfrac{b-7a}{a+5} = \dfrac{3b+a}{4b+5}$ is satisfied. Compute the value of $a + 6b$.

10. Let S be the set $\{1, 2, 3, 4\}$. Compute the number of functions $f : S \to S$ such that

$$f(1) + f(f(1)) + f(f(f(1))) + f(f(f(f(1)))) = 13.$$